

Thyristor Protection

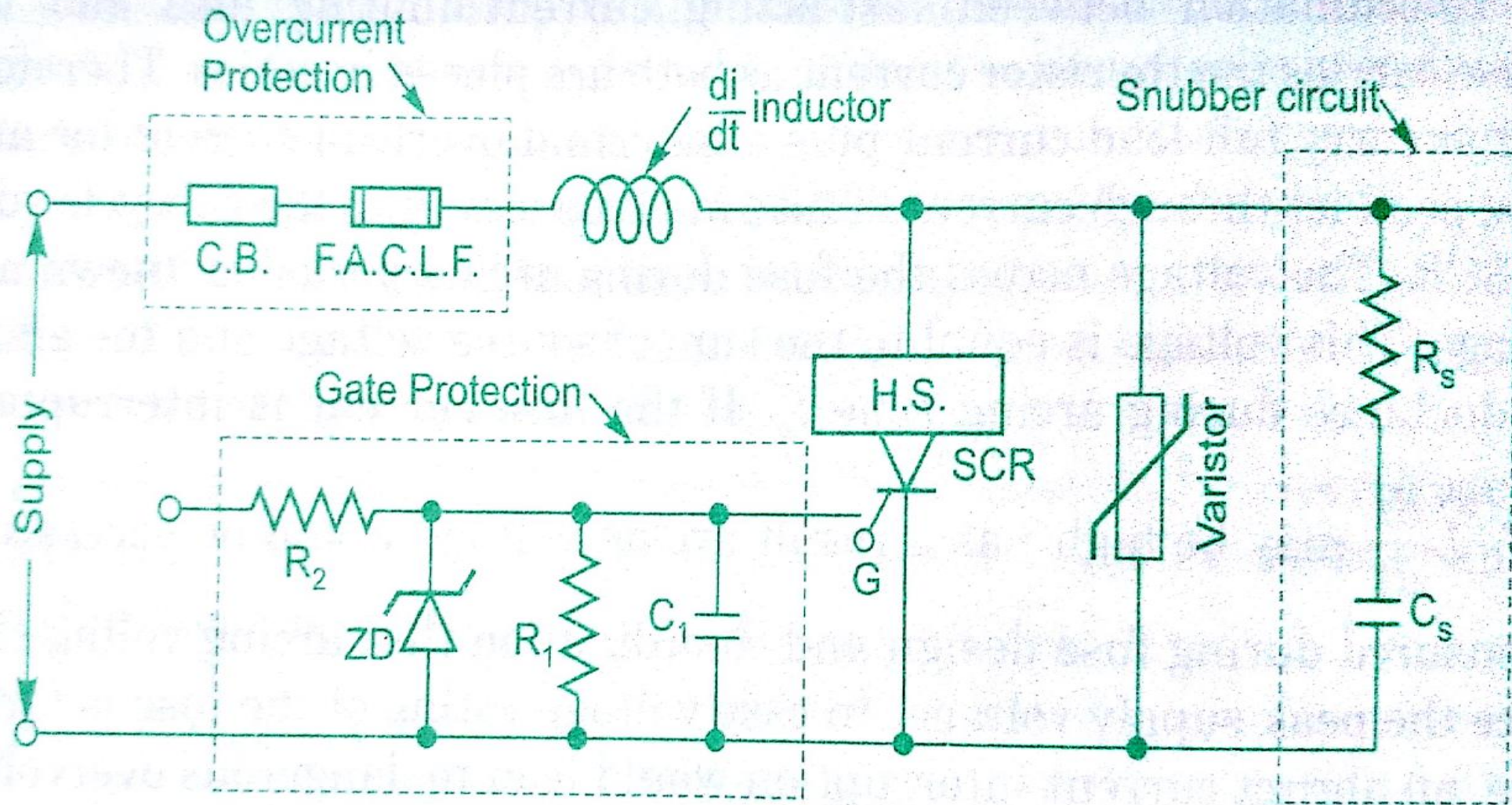
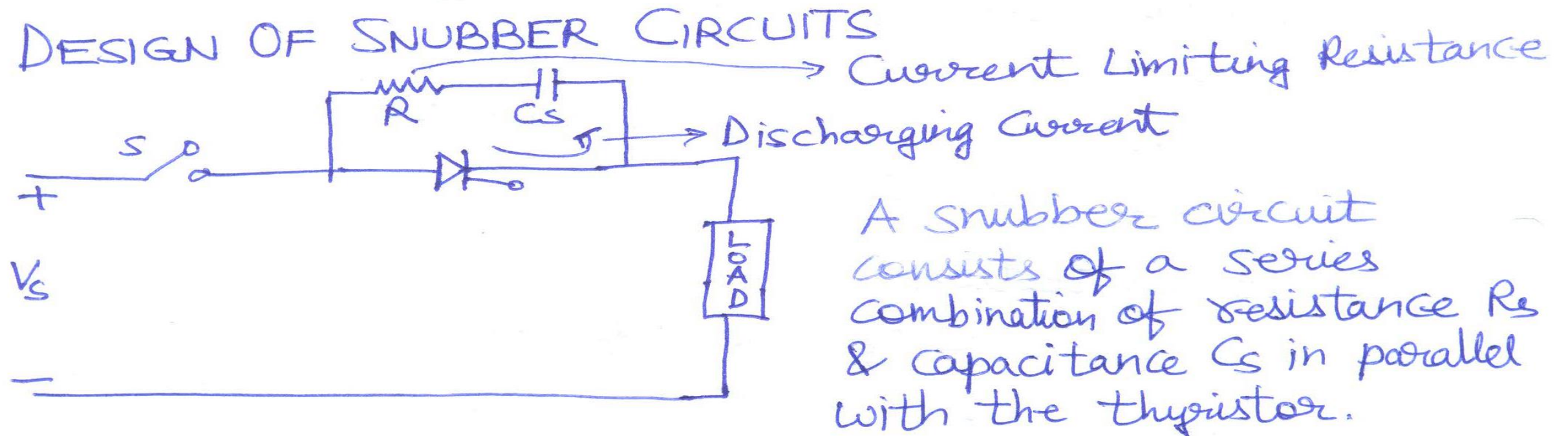


Fig. 4.30. Circuit components showing the thyristor protection.

C.B.—Circuit breaker ; F.A.C.L.F.—Fast acting current limiting fuse ; H.S.—Heat sink ; ZD—Zener diode.

DESIGN OF SNUBBER CIRCUITS



A snubber circuit consists of a series combination of resistance R_s & capacitance C_s in parallel with the thyristor.

When S is closed a sudden voltage appears across the circuit. Initially C_s behaves as a short circuit & $\therefore$ voltage across SCR is 0. Slowly, voltage across C_s builds up resulting in a value of dV/dt across SCR that is within tolerable limit.

' R ' is provided to limit the value of capacitor discharge current, once the thyristor is turned ON.

Thermal Protection

HEAT SINKS

To ensure adequate heat dissipation, heat sinks are attached to the case of the thyristor. Heat is transferred from the junction to the case, then to the sink & finally to the atmosphere.

A number of heat sinks of extruded aluminium are manufactured for use with power electronic devices. Heat sinks are cooled by natural convection. If the surface of heat sink is coated with black oxide, the thermal resistance decreases by about 25% but the cost of heat sink also increases.

For natural air cooled heat sinks the thermal time constant is in the range of 4-15 min.. In forced air cooling, the time constant is less than 1 min.. In high power devices water or air cooling is also used.

THERMAL OHM'S LAW

Heat dissipation is governed by Thermal Ohm's Law.

$$\theta_{21} = \frac{T_2 - T_1}{P_{21}} \quad \text{where } \theta_{21} \rightarrow \text{Thermal resistance between points 2 \& 1.}$$

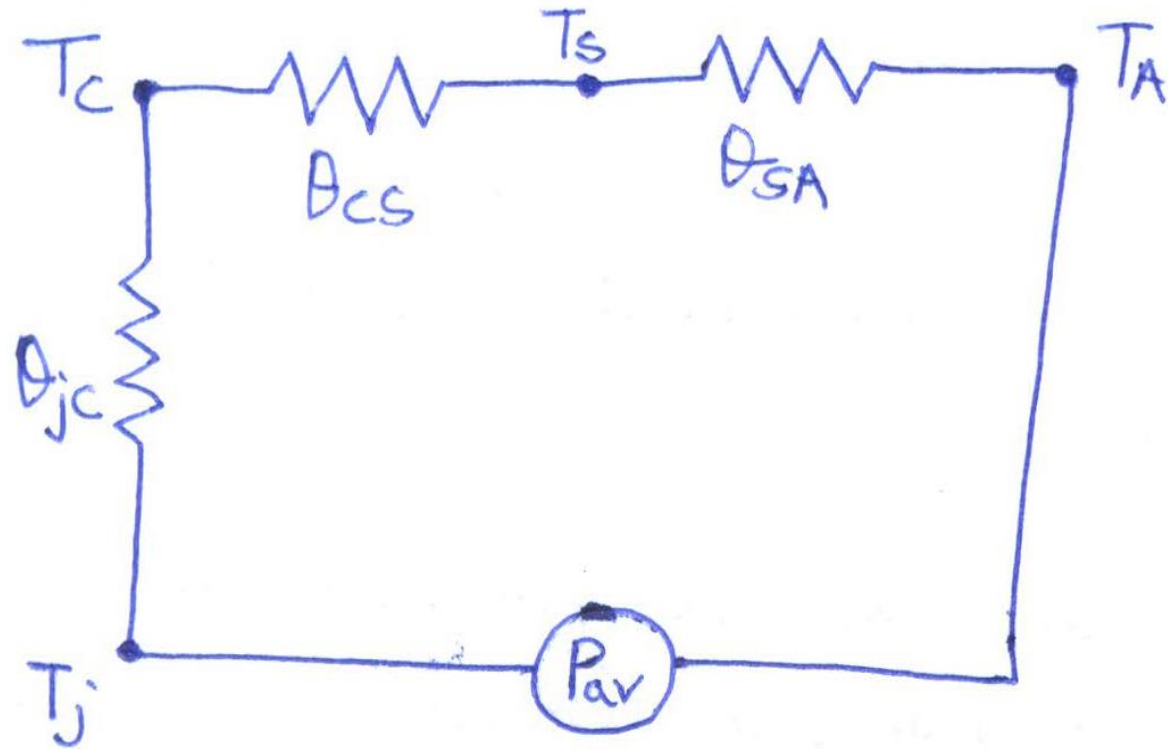
$P_{21} \rightarrow$ Power loss to be dissipated from point 2 to point 1. (W)

$T_2 \rightarrow$ Temp. of point 2 ($^{\circ}\text{C}$)

$T_1 \rightarrow$ Temp. of point 1 ($^{\circ}\text{C}$)

A thermal ohm is that thermal resistance which needs a temp. difference of 1°C across it to cause a heat flow of 1 W. ($^{\circ}\text{C}/\text{W}$)

THERMAL EQUIVALENT CIRCUIT



$P_{av} \rightarrow$ Average power loss in the device that has to be dissipated.

$T_j \rightarrow$ Temp. of junction

$T_c \rightarrow$ Temp. of case

$T_s \rightarrow$ Temp. of sink

$\theta_{jc} \rightarrow$ Thermal resistance from junction to case

$\theta_{cs} \rightarrow$ " " " Case to sink

$\theta_{sa} \rightarrow$ " " " Sink to atmosphere